Ch 2.1: What is Statistical Learning?

Lecture 2 - CMSE 381

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Announcements

Last time:

- Discussed where to find everything
- Check out the syllabus!

Announcements:

- Get on slack!
  - +1 point on the first homework if you post a gif in the thread
- First homework due Fri, Jan 21
- First lab this Friday, Jan 14
  - Be sure to have python and jupyter notebooks installed and working
- First office hours Tues, Jan 18

Covered in this class

- Input/output variables
- Prediction vs inference
- Reduceable vs irreduceable error
- Overfitting
- Classification vs regression
- Supervised vs Unsupervised learning

For next year:
They get these very confused

An example data set: `Advertising`

X_1 X_2 X_3 Y

		TV	Radio	Newspaper	Sales
1		230.1	37.8	69.2	22.1
2	1	44.5	39.3	45.1	10.4
3	2	17.2	45.9	69.3	9.3
4	3	151.5	41.3	58.5	18.5
5	4	180.8	10.8	58.4	12.9
6	5	8.7	48.9	75	7.2
7	6	57.5	32.8	23.5	11.8
8	7	120.2	19.6	11.6	13.2
9	8	8.6	2.1	1	4.8
10	9	199.8	2.6	21.2	10.6
11	10	66.1	5.8	24.2	8.6

Data available at
https://github.com/nguyen-toan/ISLR/blob/master/dataset/Advertising.csv

- Sales of a product in 200 markets, along with amount spent on three differnt types of advertising
- Goal: Predict sales based on advertising budgets
- Input variables: TV, Radio Newspaper
- Output variable: Sales

$p = 3$

Notation and Big Assumption

Input variables: $X_1, X_2, \cdots, X_p$
Output variable: Y

$p = \#$ of input variables

$f(X_1, X_2, X_3, \ldots, X_p)$

$$Y = f(X) + \varepsilon$$

Noise

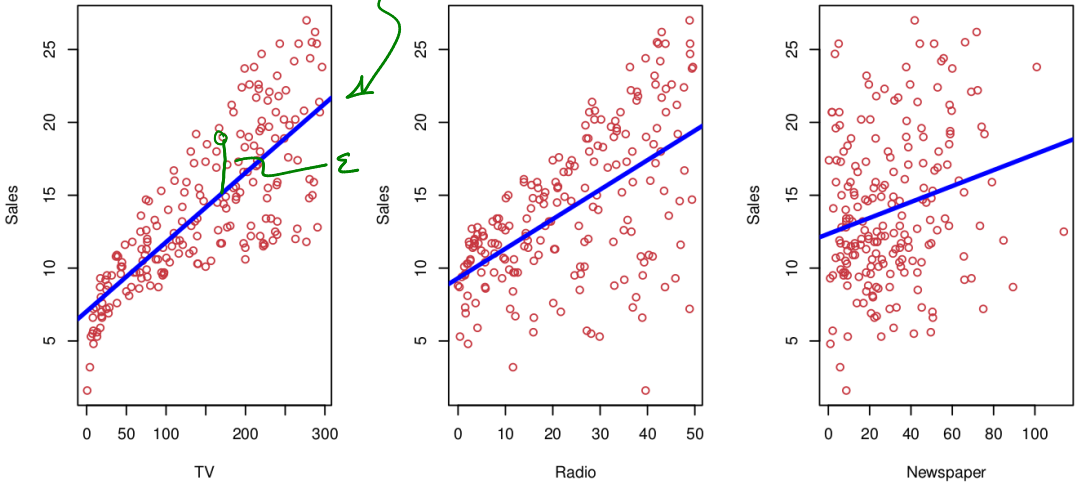
Unknowable true function

Guess that function! $\hat{f}$

Advertising Example

f = linear

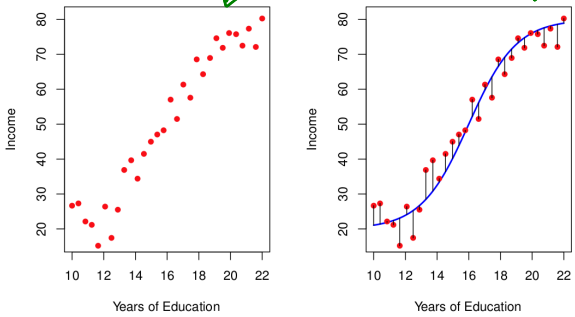
Maybe blue is estimate

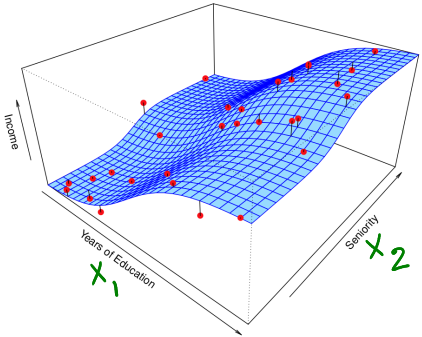


More examples

Input data

model in blue





$p = 2$

Section 1

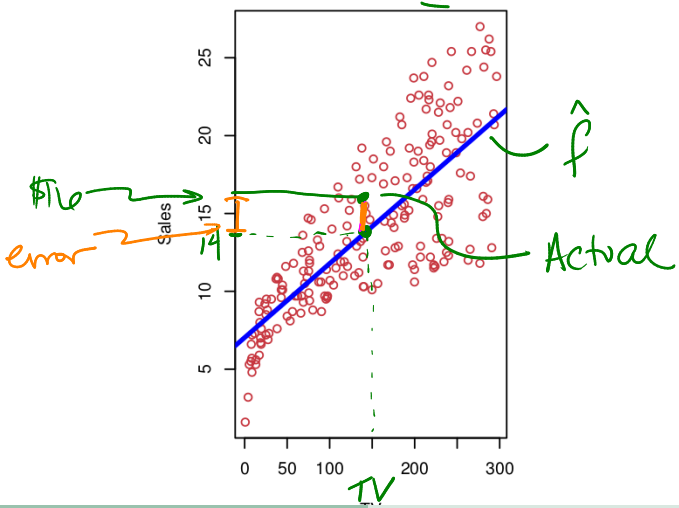
Prediction vs Inference

The question you have

Prediction

X_{TV}

Example: If we spend $150 on TV advertising, what will we make in sales?



$$\hat{Y} = \hat{f}(X)$$

$$14 = \hat{f}(150)$$

Reduceable vs irreducable error

All models are wrong, some are useful.

make this small!

Reducible Error

If I choose better
- statistical technique
- functions
- computation power

then I can make it smaller

Irreducible Error

Model $Y = f(x) + \varepsilon$

Irreduceable

More on error

not a RV!

- Given estimate $\hat{f}$ (fixed)
- Set of predictors X (fixed) — Not a RV
- Prediction $\hat{Y} = \hat{f}(X)$

$$Y = f(X) + \varepsilon$$
$$\hat{Y} = \hat{f}(X)$$

$$E(Y - \hat{Y})^2 = Var(Y - \hat{Y}) + [E(Y - \hat{Y})]^2$$

expected value

$$= Var(f(X) + \varepsilon - \hat{f}(X)) + [E(f(X) + \varepsilon - \hat{f}(X))]^2$$

Assume $E(\varepsilon) = 0$

$$E(Z^2) = Var(Z) + [E(Z)]^2$$

$$= \underbrace{Var(\varepsilon)}_{\text{Irreducable}} + \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{reducable}}$$

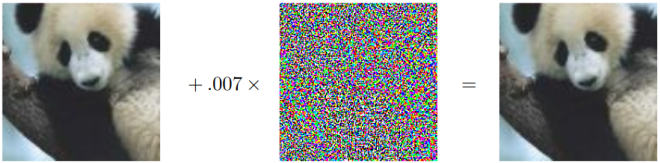
Inference

Want f, but not for prediction (or possibly combined with prediction)

- Which predictors are associated with the response?
- What is the relationship between the response and each predictor?
- Can the relationship between Y and each predictor be adequately summarized using a linear equation? Is it more complicated?

Inference is important!

noise



Self-driving cars

Images from: arxiv.org/abs/1707.08945, arxiv.org/abs/1412.6572

Determine whether each scenario is prediction, inference, or both.

Application	Prediction	Inference
Identify risk of getting covid	X	X — what comorbidities?
Predict effectiveness of vaccine	X	X
Recognize handwritten text	X	?
Speech recognition	X	?
Predict stock prices	X	?

Section 2

How to estimate f?

Input: Training data

$\cancel{p=7}$ $p=17$ $p=3$

$x_8 = (x_{81}, x_{82}, x_{83}, \cdots, x_{817})$

X_1 X_2 X_3 Y

- n data points observed
- x_{ij} is the jth predictor for observation i
- y_i is the response variable for the ith observation
- Training data:
  - $\{(x_1, y_1), (x_2, y_2), \cdots, (x_n, y_n)\}$
  - $x_i = (x_{i1}, x_{i2}, \cdots, x_{ip})^T$

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$x_{5,2}$ y_5 $n=300$ x_8

Parametric methods

Step 1: Select a model

parameters

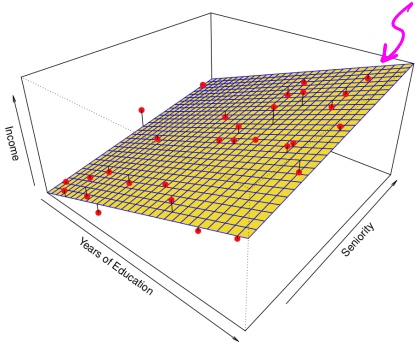
Example:

$$f(X) = \beta_0 + \beta_1 X_2 + \beta_2 X_2 + \cdots + \beta_p X_p$$

Step 2: Train the model

Example:
Find $\beta_i's$ so that

$$Y \approx \beta_0 + \beta_1 X_2 + \beta_2 X_2 + \cdots + \beta_p X_p$$

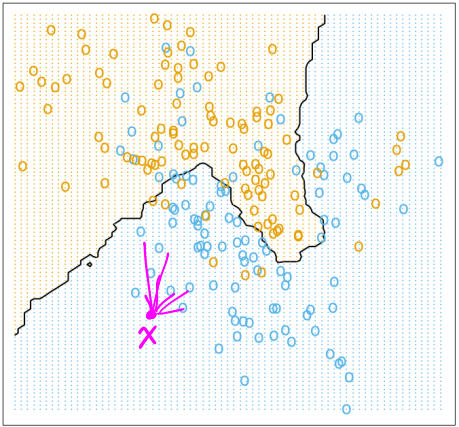


Example Non-parametric method: Nearest Neighbors

hyper parameter

$$N_k(x) = \text{Set of } k \text{ nearest neighbors of } x$$

$$\hat{f}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$



$k = 15$

Parametric methods: Pros and Cons

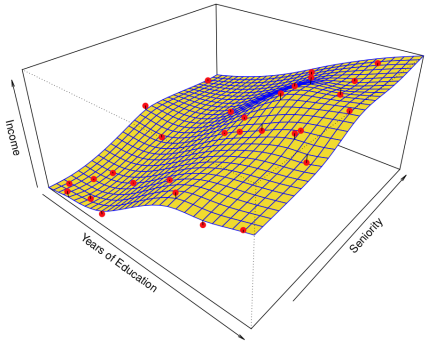
Pros

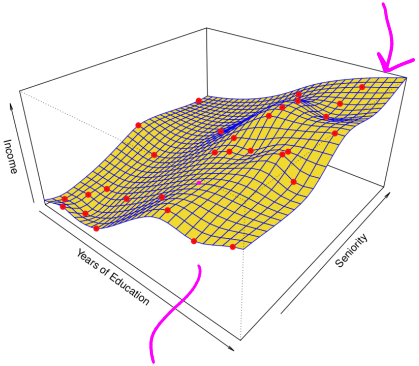
Easier to estimate parameters than to decide on a completely arbitrary function

Cons

- Could be the "wrong" model/function to use
- Very limiting

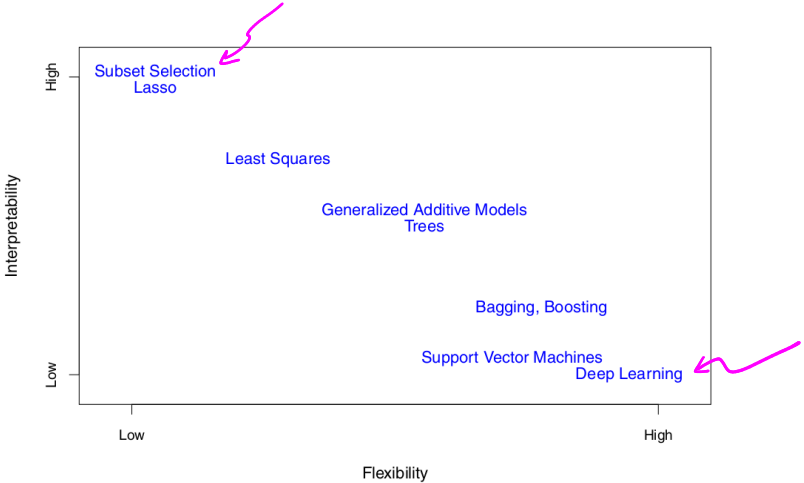
Overfitting





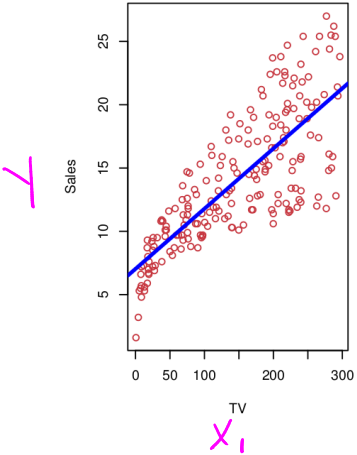
zero error according to data

Prediction Accuracy vs Model Interpretability

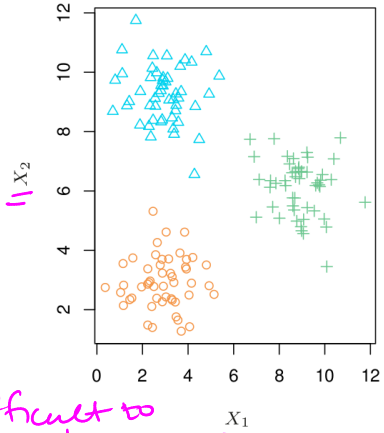


Supervised learning: Training data has response variable y for every input x

Unsupervised Learning: Training data has ~~response~~ does not! variable y for every input x



(semi supervised learning)



Difficult to decide how well you did?

Regression vs Classification

For next year: More emphasis on this being output variable

Types of variables:

- Quantitative
  - ↳ value in $\mathbb{R}$
  - (Blood pressure, Income; Temperature, Height; Humidity, Volume)
- Qualitative / Categorical
  - ↳ Elements of a set / list
    - Names
    - Owns a house
    - College degree

→ Regression problem

→ Classification problem

Section 3

Group work

data pts

(a) We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.

Y

salary

- Is this classification or regression?
- Do we want inference or prediction?
- What is n, the number of data points? 500
- What is p, the number of variables?

$p=3$

input

$X_1, \ldots, X_p$ $\quad$ Y

(b) We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

$\sim Y$

- Is this classification or regression?
- Do we want inference or prediction?
- What is n, the number of data points? 20
- What is p, the number of variables?

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Wrap up

Next time:

- Friday:
  - Lab, be sure to install python/jupyter!
  - No homework due until next week
- ~~Monday:~~ Wednesday
  - Ch 2.2: Assessing Model Accuracy

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